

2022 HKDSE Physics

Paper 2 — Elective Sections

Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

Editable questions with official answers and detailed English reasoning. Subscripts and superscripts remain editable.

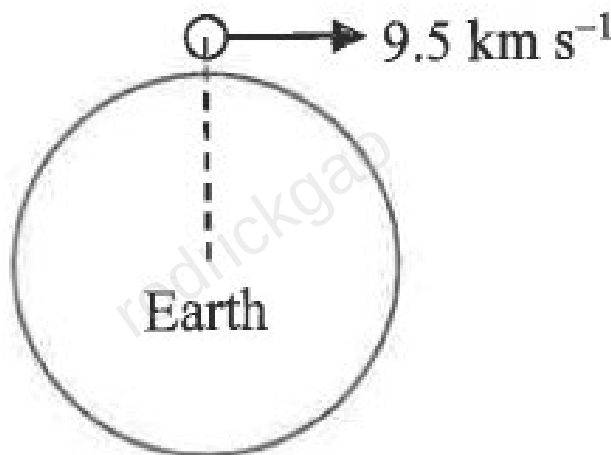
Section A — Astronomy and Space Science

Question 1: Multiple-choice questions

Official answers are placed directly below each question.

1.1

The minimum velocity for an object to keep orbiting the Earth is 7.9 km s^{-1} , whereas the escape velocity is 11.2 km s^{-1} . What kind of orbit would an object travel if it is launched horizontally with a velocity of 9.5 km s^{-1} ? Assume that atmospheric drag is negligible.



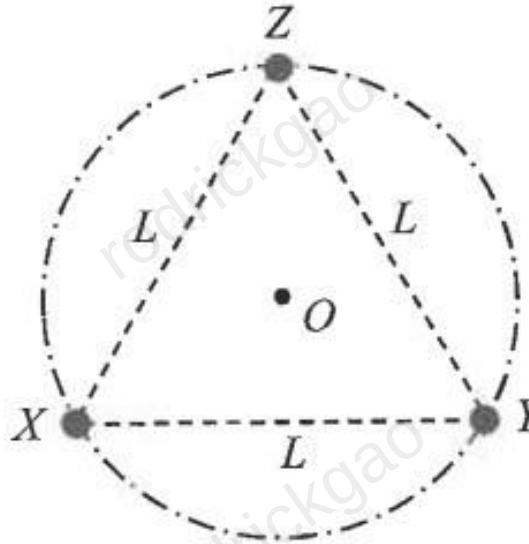
- A. circular orbit
- B. elliptical orbit
- C. parabolic orbit
- D. along a straight line

Official answer: B

The launch speed lies above the circular-orbit speed but below escape speed. The object remains gravitationally bound, with its launch point at the perigee of an elliptical orbit. Answer: B.

1.2

Three identical stars X, Y and Z are situated at the vertices of an equilateral triangle of side length L . They perform uniform circular motions with angular velocity ω about the centre O of the triangle. Which of the following is the relationship between ω and L ?



- A. $\omega \propto \sqrt{(1/L^3)}$
- B. $\omega \propto \sqrt{(L^3)}$
- C. $\omega \propto \sqrt{(1/L)}$
- D. $\omega \propto \sqrt{L}$

Official answer: A

For a star of mass m , the resultant attraction is $\sqrt{3}Gm^2/L^2$ towards O . The orbital radius is $L/\sqrt{3}$, so $m\omega^2(L/\sqrt{3}) = \sqrt{3}Gm^2/L^2$. Hence $\omega^2 = 3Gm/L^3$ and $\omega \propto L^{-3/2}$. Answer: A.

1.3

A satellite initially at rest is launched from the Earth's surface into an orbit at a height R above the Earth's surface in which the satellite performs uniform circular motion, where R is the radius of the Earth. The gain in kinetic energy of the satellite in this process is K . What is the corresponding change in the satellite's gravitational potential energy?

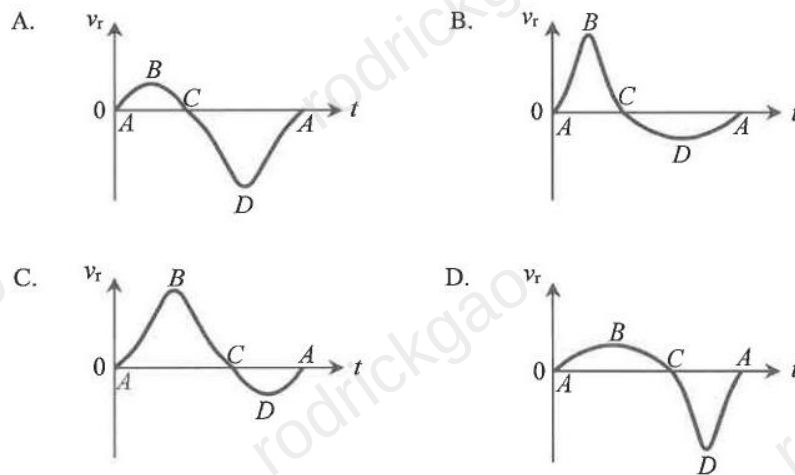
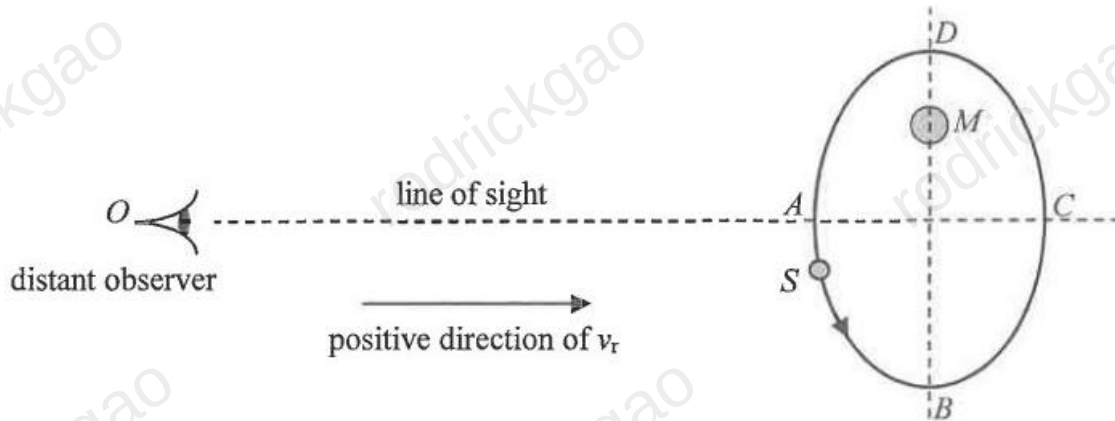
- A. $-2K$
- B. $-K$
- C. $+2K$
- D. $+K$

Official answer: C

The final orbital radius is $2R$. Circular motion gives $K = GMm/[2(2R)] = GMm/(4R)$. The potential-energy increase is $\Delta U = -GMm/(2R) - (-GMm/R) = GMm/(2R) = +2K$. Answer: C.

1.4

In the binary star system shown below, star M is a massive star which is nearly stationary. Star S moves around M in an elliptical orbit with major axis BD. A distant observer O is in the orbital plane of star S, with the line of sight perpendicular to BD. Which graph best shows the variation of the observed radial velocity v_r of S with time t ? S travels anticlockwise A–B–C–D–A and its positions A, B, C and D are marked correspondingly on the graphs.



Official answer: D

At B the star recedes from O, giving positive radial velocity; at D it approaches, giving negative radial velocity. D is closest to the massive star, so the negative peak is larger in magnitude and narrower in time. The line-of-sight component need not vanish exactly at A or C because the force centre M lies off the line AC. Graph D has the correct signs and unequal peaks. Answer: D.

1.5

In the same star cluster, a blue star and a red star have the same apparent magnitude. What is the ratio radius of blue star: radius of red star if the surface temperature of the blue star is twice that of the red star?

- A. 1: 4
- B. 1: 2
- C. 2: 1
- D. 4: 1

Official answer: A

The two stars are at approximately the same distance and have equal apparent brightness, so their luminosities are equal. From $L = 4\pi R^2 \sigma T^4$, $R \propto 1/T^2$ for fixed L . Therefore $R_{\text{blue}}/R_{\text{red}} = (1/2)^2 = 1/4$. Answer: A.

1.6

Two distant stars X and Y have the same absolute magnitude but X appears 4 times as bright as Y to the naked eye. What is the ratio of the stellar parallax of X to that of Y?

- A. 1: 4
- B. 1: 2
- C. 2: 1
- D. 4: 1

Official answer: C

Equal absolute magnitudes mean equal luminosities. Since brightness is proportional to $1/d^2$, $b_X/b_Y = 4$ gives $d_Y/d_X = 2$. Parallax is inversely proportional to distance, so $p_X:p_Y = 2:1$. Answer: C.

1.7

The calcium H line (396.8 nm) from a distant object appears to be at 395.8 nm in the absorption spectrum observed. From this information, one can conclude that the object moves

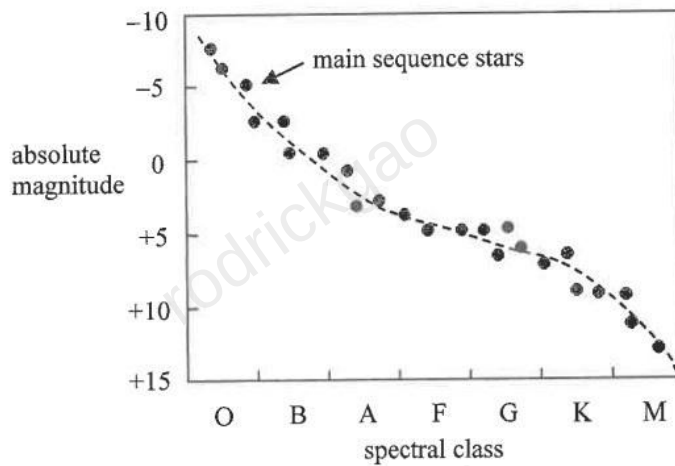
- A. towards the Earth at a speed of 756 km s^{-1} .
- B. away from the Earth at a speed of 756 km s^{-1} .
- C. towards the Earth at a speed of at least 756 km s^{-1} .
- D. away from the Earth at a speed of at least 756 km s^{-1} .

Official answer: C

The shorter observed wavelength is a blueshift, so the radial motion is towards Earth. Its magnitude is $|v_r| = c|\Delta\lambda/\lambda_0| = (3.00 \times 10^5)(1.0/396.8) = 756 \text{ km s}^{-1}$. Doppler shift measures only the radial component, so the total speed is at least this value. Answer: C.

1.8

The Hertzsprung–Russell (H–R) diagram below shows a series of main sequence stars. The properties of three main sequence stars X, Y and Z are tabulated below. Which of them is/are farther than 10 pc from the Earth?



- A. X only
- B. Y only
- C. X and Z only
- D. Y and Z only

Star	Apparent magnitude	Spectral class
X	0	F
Y	5	B
Z	4	K

Official answer: B

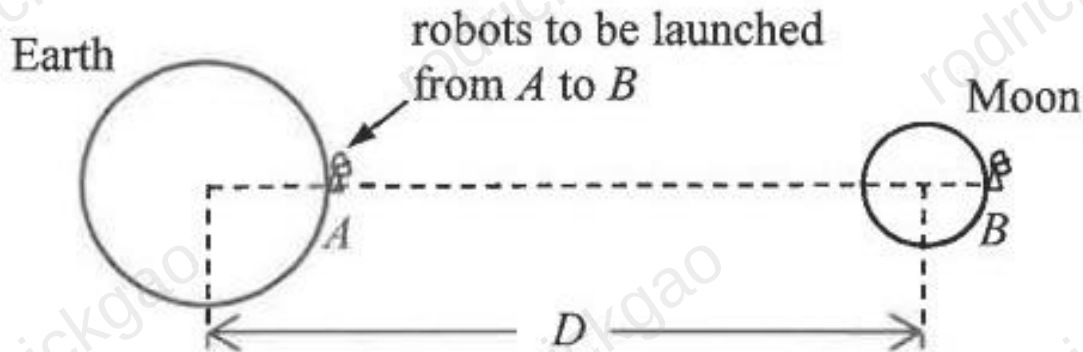
At 10 pc, apparent magnitude m equals absolute magnitude M . A star is farther than 10 pc when $m > M$.

Reading the main sequence, the B-class star Y has M well below its $m = 5$, while X and Z have m smaller than their corresponding M . Only Y is farther than 10 pc. Answer: B.

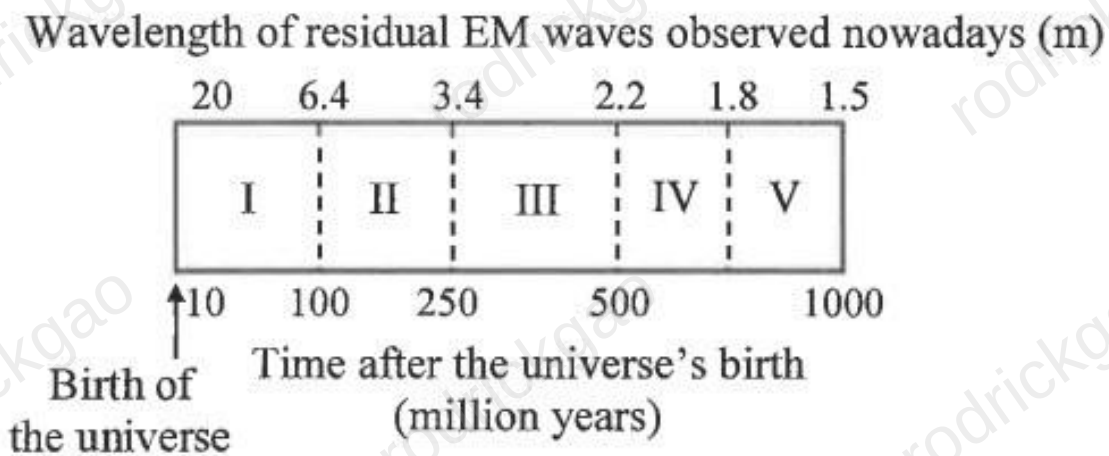
Question 1: Structured question

Given: mass of the Moon = $0.0123 \times$ mass of the Earth; radius of the Moon = $0.273 \times$ radius of the Earth.

- (a) Based on Newton's universal law of gravitation, estimate the ratio of gravitational acceleration on the Moon's surface, g_M , to that on the Earth's surface, g_E . Give your answer correct to 3 significant figures. (2 marks)
- (b) Scientists plan to deploy robots to the far side of the Moon to build radio telescopes for observing electromagnetic waves (EM waves).



- (i) At a certain point N between the Earth and the Moon, the gravitational forces due to the Earth and the Moon balance each other. Estimate the distance of N from the Earth's centre in terms of the average Earth–Moon separation D . (2 marks)
- (ii) For launching an object from point A on the Earth's surface so that it eventually reaches the Moon, the object must possess enough energy to reach point N. Estimate the minimum launching velocity of the robot. Given: the change in gravitational potential energy of an object from A to N in the Earth–Moon system is $6.12 \times 10^7 \text{ J kg}^{-1}$. (2 marks)
- (c) Our universe is estimated to be 14 billion (i.e. 14 000 million) years old. Scientists discovered that hydrogen atoms in the early universe (i.e. 0.4 to 1000 million years since its birth) have been emitting EM waves of wavelength 21 cm. As the universe has been expanding over time, these residual EM waves observed nowadays have their wavelengths stretched to different extents depending on when the waves were produced (see Figure 1.2).



- Name the phenomenon of the 'stretching' of wavelengths mentioned above. (1 mark)
- The radio telescope in (b) covers a range of frequencies including $f_c = 20$ MHz. Determine the wavelength λ_c corresponding to f_c and name this kind of EM wave. (2 marks)
- Referring to Figure 1.2, identify the period (I to V) that the EM waves in (c)(ii) come from. (1 mark)

Official marking scheme

Solution	Marks	Remarks
1. (a) Gravitational acceleration on a celestial body's surface is given by $g = \frac{GM}{R^2} \quad \left(\text{as } F = \frac{GMm}{R^2} = mg \right)$ $\therefore \frac{g_M}{g_E} = \left(\frac{M_{Moon}}{M_{Earth}} \right) \left(\frac{R_{Earth}}{R_{Moon}} \right)^2$ $= (0.0123) \left(\frac{1}{0.273} \right)^2 = 0.165036 \approx 0.165 \text{ (3 sig. fig.)}$	1M 1A 2	Accept 0.164 ~ 0.166
(b) (i) Let r be the distance of N from the Earth's centre, $\frac{GM_{Earth}m}{r^2} = \frac{GM_{Moon}m}{(D-r)^2} \quad \text{or} \quad \frac{GM_{Earth}}{r^2} = \frac{GM_{Moon}}{(D-r)^2}$ $\frac{D-r}{r} = \sqrt{\frac{M_{Moon}}{M_{Earth}}} = \sqrt{0.0123}$ $r = \frac{1}{1+\sqrt{0.0123}} D = 0.900167 D \approx 0.90 D$	1M 1A 2	Accept 0.89 D ~ 0.91 D
(ii) The required kinetic energy = increase in gravitational potential energy $\frac{1}{2}mv^2 = m(6.12 \times 10^7)$ $v = 11063.453 \text{ m s}^{-1} \approx 11.1 \text{ km s}^{-1}$	1M 1A 2	Accept 11.0 km s^{-1} ~ 11.2 km s^{-1}
(c) (i) redshift	1A 1	
(ii) $v = f\lambda$ $3 \times 10^8 = 20 \times 10^6 \lambda_c$ $\lambda_c = 15 \text{ m}$ radio waves / short waves	1M/1A 1A 2	
(iii) Period: I	1A 1	

(a) With $g = GM/R^2$, $g_M/g_E = 0.0123/(0.273^2) = 0.165$ to three significant figures.

(b)(i) Let r be the distance from Earth's centre. Equal attractions require $GM_E/r^2 = GM_M/(D-r)^2$. Thus $(D-r)/r = \sqrt{0.0123}$ and $r = D/(1+\sqrt{0.0123}) = 0.900D$.

(b)(ii) For the minimum launch speed, $\frac{1}{2}mv^2 = m(6.12 \times 10^7 \text{ J kg}^{-1})$. Therefore $v = \sqrt{(2 \times 6.12 \times 10^7)} = 1.11 \times 10^4 \text{ m s}^{-1} = 11.1 \text{ km s}^{-1}$.

(c)(i) The phenomenon is cosmological redshift.

(c)(ii) $\lambda_c = c/f_c = 3.00 \times 10^8 / (20 \times 10^6) = 15 \text{ m}$; these are radio waves.

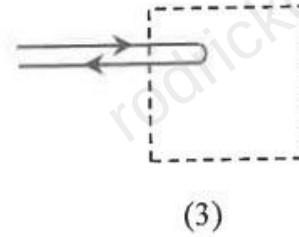
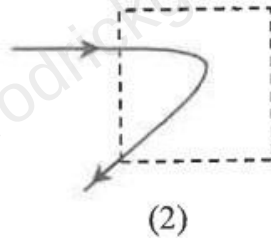
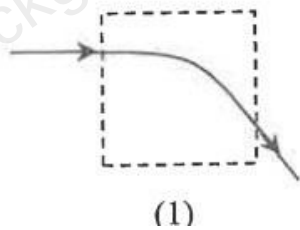
(c)(iii) Period I: 15 m lies between the 20 m and 6.4 m boundaries in Figure 1.2.

Section B — Atomic World

Question 2: Multiple-choice questions

2.1

Each figure shows the path of an α particle scattered by a fixed charged particle P (not shown but inside the dotted region). If electrostatic force is the only interaction between the two particles, from which figure can one deduce that P must be positively charged?



- A. (3) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

In figure (3), the α -particle is strongly backscattered, turning back before passing the scattering region. This requires a repulsive interaction, so P must be positive. Curved scattering paths (1) and (2) alone do not determine the sign when the position of P is unknown; attraction towards a different position can also produce a bend. Answer: A.

2.2

Which statements about the absorption spectrum of helium are correct?

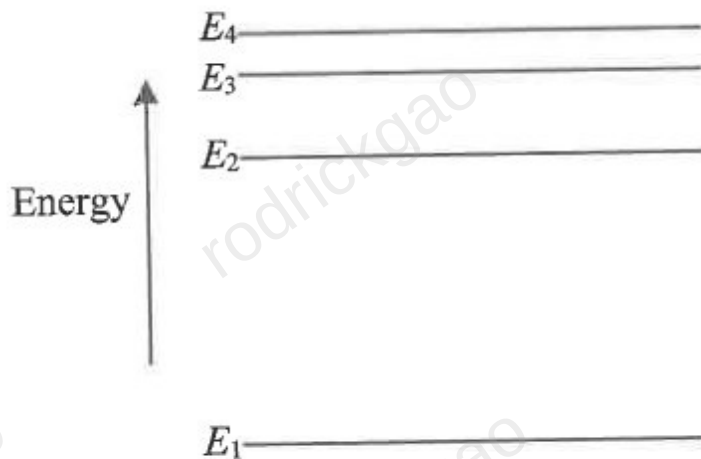
- (1) Helium atoms can only absorb photons with energy that equals the difference between two energy levels of helium.
 - (2) The dark lines in the absorption spectrum of helium match those in its emission spectrum.
 - (3) The dark lines in the spectrum correspond to the wavelength of photons being absorbed by helium atoms.
- A. (1) and (2) only
 - B. (1) and (3) only
 - C. (2) and (3) only
 - D. (1), (2) and (3)

Official answer: D

For the bound-state spectral lines considered, a photon is absorbed when its energy equals an allowed energy-level difference. The missing wavelengths form dark absorption lines and coincide with the corresponding helium emission lines. All three statements are correct. Answer: D.

2.3

The diagram represents the energy levels drawn to scale for an electron in a certain atom. The electron transition from E_3 to E_1 produces a green line. What electron transition would give rise to a red line? Given: the visible spectrum is about 400 nm to 750 nm.



- A. E_3 to E_2
- B. E_4 to E_2
- C. E_2 to E_1
- D. E_4 to E_1

Official answer: C

Red photons have less energy than green photons because $E = hc/\lambda$. The scaled diagram shows that $E_2 \rightarrow E_1$ has a moderately smaller energy gap than the green $E_3 \rightarrow E_1$ transition, consistent with red light. The gaps $E_3 \rightarrow E_2$ and $E_4 \rightarrow E_2$ are much smaller and lie outside the visible range on the long-wavelength side. Answer: C.

2.4

According to the Bohr model of the hydrogen atom, which physical quantities about the electron decrease when a photon is emitted?

- (1) orbital radius
- (2) angular momentum
- (3) kinetic energy

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Emission moves the electron to a lower Bohr level n . Both $r \propto n^2$ and angular momentum $nh/(2\pi)$ decrease, but kinetic energy is proportional to $1/n^2$ and increases. Statements (1) and (2) are correct. Answer: A.

2.5

An optical microscope is employed to view a specimen such that two points of separation 3×10^{-6} m can just be resolved. If the diameter of the objective lens is 10 mm, deduce a probable combination of objective-to-specimen separation L and wavelength λ of the light used.

- A. $L = 5$ cm; $\lambda = 410$ nm
- B. $L = 4$ cm; $\lambda = 615$ nm
- C. $L = 3$ cm; $\lambda = 410$ nm
- D. $L = 2$ cm; $\lambda = 615$ nm

Official answer: B

At the resolution limit, $s/L = 1.22\lambda/d$. For $s = 3 \times 10^{-6}$ m and $d = 0.010$ m, the required product λL is $sd/1.22 = 2.46 \times 10^{-8}$ m². Option B gives $(615 \times 10^{-9})(0.040) = 2.46 \times 10^{-8}$ m². Answer: B.

2.6

A voltage V_0 is applied to the anode of a transmission electron microscope (TEM) to produce an electron beam of de Broglie wavelength 0.01 nm. What should the anode voltage be for reducing the wavelength to 0.005 nm?

- A. $V_0/4$
- B. $V_0/2$
- C. $2V_0$
- D. $4V_0$

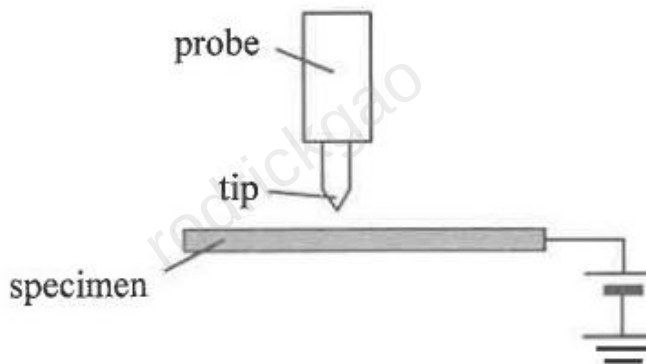
Official answer: D

For a non-relativistic electron accelerated through V , $eV = p^2/(2m)$ and $\lambda = h/p$, so $\lambda \propto V^{-1/2}$. Halving the wavelength requires four times the accelerating voltage: $4V_0$. Answer: D.

2.7

The simplified schematic diagram below shows a scanning tunnelling microscope (STM). Which statements about its operation are correct?

- (1) The tunnelling current increases with the conductivity of the specimen's surface.
- (2) The tunnelling current is independent of the medium between the tip and the specimen.
- (3) When scanning in constant-height mode, the change in distance between the tip and the specimen's surface is in atomic scale.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

The conducting surface supplies the tunnelling electrons, so its conductivity affects the current. Atomic-scale changes of tip-to-surface distance cause strong current changes in constant-height operation. The medium affects the tunnelling barrier, so the current is not independent of it. Statements (1) and (3) are correct. Answer: C.

2.8

Which of the following is/are in nanoscale?

- (1) the diameter of a human hair
- (2) the size of a gold nucleus
- (3) the size of a coronavirus (e.g. COVID-19) particle

A. (1) only

B. (3) only

C. (1) and (2) only

D. (2) and (3) only

Official answer: B

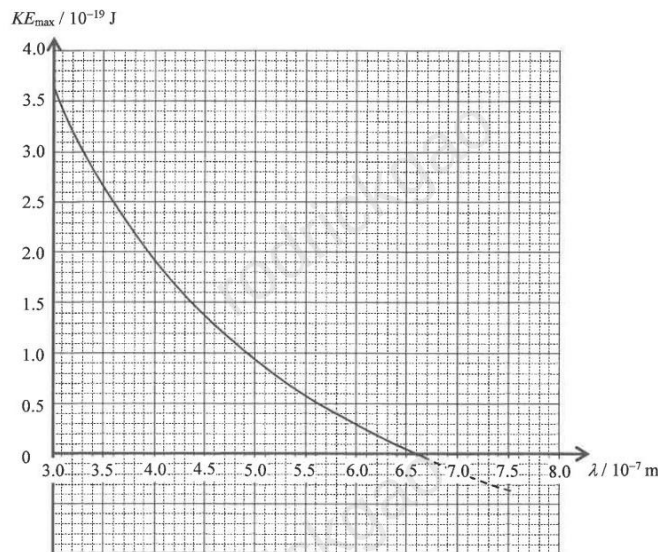
A human hair is typically tens of micrometres across, much larger than the nanoscale. A gold nucleus is only a few femtometres across, much smaller. A coronavirus particle is of order 10^2 nm, the nanoscale object among those listed. Answer: B.

Question 2: Structured question

When monochromatic light is incident on a metal surface, electrons within a certain range of energies are emitted. Each photon of the monochromatic light carries 4.97×10^{-19} J of energy.

- (a) (i) Explain whether the range mentioned above would be affected if the intensity of this monochromatic light is increased. (2 marks)
- (ii) Given that the light intensity is 0.050 W m^{-2} and the area of the metal surface is $1.00 \times 10^{-4} \text{ m}^2$, estimate the maximum number of electrons emitted per second. (2 marks)

The metal surface is illuminated by monochromatic lights of different wavelengths λ in turn. The corresponding maximum kinetic energies K_{max} of the emitted electrons are obtained by measuring the stopping potential V_s . A graph of K_{max} against λ is plotted as shown in Figure 2.1.



- (b) If monochromatic light of wavelength $4.0 \times 10^{-7} \text{ m}$ is used, find the corresponding stopping potential. (2 marks)
- (c) Find the work function, in eV, of the metal. (3 marks)
- (d) If another metal with a larger work function is used, would the intercept on the vertical axis increase, decrease or remain unchanged? (1 mark)

Official marking scheme

Solution		Marks	Remarks
2. (a)	(i) As wavelength / frequency / energy of photon remains unchanged, the same maximum kinetic energy $KE_{\max}$ for the electrons emitted. The range does not depend on the intensity of incident light, i.e. remains unchanged.	1A	Accept $1.00 \times 10^{13} \sim 1.01 \times 10^{13}$ Accept 1.87 (eV) ~ 1.93 (eV)
		1A	
		2	
	(ii) Assuming all incident photons cause emission of electrons, maximum no. of electrons emitted in 1 second $= \frac{(0.050)(1.00 \times 10^{-4})}{4.97 \times 10^{-19}} = 1.00603622 \times 10^{13}$ $\approx 1.01 \times 10^{13}$	1M	
		1A	
		2	
	(b) $KE_{\max} = eV_s$ (or $q_e V_s$) $1.9 \times 10^{-19} = eV_s$ $V_s = \frac{1.9 \times 10^{-19}}{1.60 \times 10^{-19}}$ $= 1.1875 \text{ V} \approx 1.19 \text{ V}$	1M	
		1A	
		2	
	(c) Threshold wavelength $\lambda_0 = 6.6 \times 10^{-7} \text{ m}$ (from graph) Work function $= \frac{hc}{\lambda_0} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{6.6 \times 10^{-7}}$ $= 3.0136 \times 10^{-19} \text{ J}$ $= \frac{3.0136 \times 10^{-19}}{1.60 \times 10^{-19}} = 1.88 \text{ (eV)}$	1M	
		1M	
		1A	
	Or Work function $= \frac{hc}{\lambda} - KE_{\max}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{4 \times 10^{-7}} - 1.9 \times 10^{-19}$ $= 1.92 \text{ (eV)}$	1M	
		1M	
		1A	
		3	
	(d) decrease	1A	
		1	

(a)(i) The energy range remains unchanged. Increasing intensity increases the number of incident photons per second, but the photon energy hf is unchanged; consequently $K_{\max} = hf - \phi$ is unchanged.

(a)(ii) At most one emitted electron is produced per incident photon. The maximum emission rate is $IA/E_{\text{photon}} = 0.050(1.00 \times 10^{-4})/(4.97 \times 10^{-19}) = 1.01 \times 10^{13} \text{ s}^{-1}$.

(b) At $\lambda = 4.0 \times 10^{-7} \text{ m}$, read $K_{\max} \approx 1.9 \times 10^{-19} \text{ J}$. Since $eV_s = K_{\max}$, $V_s = 1.9 \times 10^{-19}/(1.60 \times 10^{-19}) = 1.19 \text{ V}$.

(c) Read the threshold wavelength $\lambda_0 \approx 6.6 \times 10^{-7} \text{ m}$. Then $\phi = hc/\lambda_0 = 3.01 \times 10^{-19} \text{ J} = 1.88 \text{ eV}$. Alternatively, $\phi = hc/\lambda - K_{\max}$ at the point in part (b) gives about 1.92 eV; the difference is within the graph-reading tolerance.

(d) It decreases. At any fixed wavelength, $K_{\max} = hc/\lambda - \phi$ decreases when ϕ increases. The displayed vertical axis is at $\lambda = 3.0 \times 10^{-7} \text{ m}$, so its intersection with the curve moves downward; it is not an intercept at zero wavelength.

Section C — Energy and Use of Energy

Question 3: Multiple-choice questions

3.1

Three types of lamps P, Q and R are available. When they operate at their respective ratings under the same conditions, which one is the brightest and which one consumes the least amount of energy?

- A. brightest Q; least energy P
- B. brightest Q; least energy Q
- C. brightest R; least energy P
- D. brightest R; least energy Q

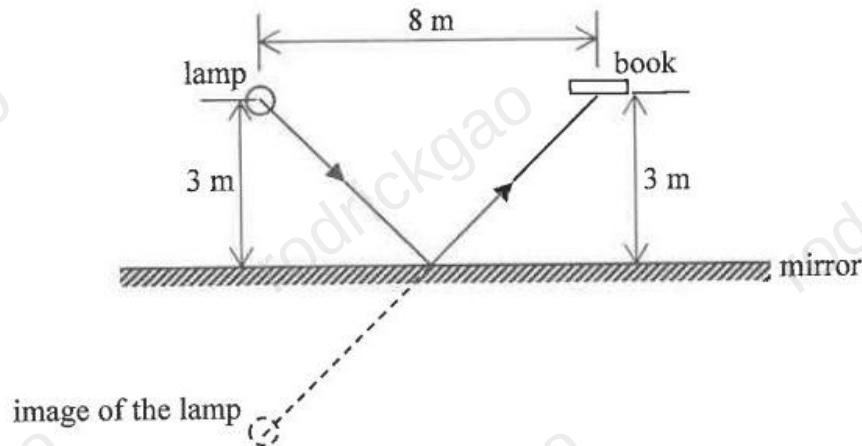
Lamp	Power rating / W	Efficacy / lm W^{-1}
P	6	100
Q	8	110
R	10	85

Official answer: A

Luminous flux equals efficacy times input power. The fluxes are P: $6(100) = 600 \text{ lm}$, Q: $8(110) = 880 \text{ lm}$ and R: $10(85) = 850 \text{ lm}$. Q is brightest, while P uses the least energy over the same operating time. Answer: A.

3.2

A lamp of luminous flux 6000 lm is placed 3 m above a horizontal mirror, and a book is placed horizontally at the same height as the lamp. Light from the lamp is reflected by the mirror to the book. If the mirror reflects 100% of the light energy, find the illuminance on the side of the book facing the mirror.



- A. 2.9 lx
- B. 3.8 lx
- C. 4.8 lx
- D. 11.5 lx

Official answer: A

Replace the reflected path by a straight path from the lamp's virtual image, 3 m below the mirror. The image-to-book distance is $\sqrt{8^2 + 6^2} = 10 \text{ m}$ and $\cos\theta = 6/10$ for the book's downward normal. Illuminance $E = [6000/(4\pi \times 10^2)](6/10) = 2.86 \text{ lx} \approx 2.9 \text{ lx}$. Answer: A.

3.3

The upper and lower reservoirs of a hydroelectric power plant have a height difference of 80 m in water levels. The turbine gives a total output power of 1000 MW . Assuming an overall efficiency of 80% , estimate the water flow rate, in kg s^{-1} , through the turbine. ($g = 9.81 \text{ m s}^{-2}$)

- A. 1.02×10^6
- B. 1.27×10^6
- C. 1.52×10^6
- D. 1.59×10^6

Official answer: D

The output power is $P = \eta \dot{m}gh$. Thus $\dot{m} = 1000 \times 10^6 / [0.80(9.81)(80)] = 1.59 \times 10^6 \text{ kg s}^{-1}$. Dividing by efficiency is necessary because the gravitational input power exceeds the useful output. Answer: D.

3.4

According to the Energy Efficiency Labelling Scheme (EELS), which statement about an air-conditioner of grade 1 is correct?

- A. It is more energy efficient than a refrigerator of grade 5.
- B. Its electric power consumption must be less than that of an air-conditioner of grade 5.
- C. Its energy efficiency must be above the average energy efficiency of air-conditioners in the same category of cooling capacity.
- D. Its energy efficiency must be below the average energy efficiency of air-conditioners in the same category of cooling capacity.

Official answer: C

Under the grading framework stated in this 2022 question, grade 1 identifies high energy efficiency within the relevant cooling-capacity category. It does not imply lower absolute power consumption than every other air-conditioner, nor permit comparison with a refrigerator. The correct within-category comparison is C. Answer: C.

3.5

The COP (coefficient of performance) of an air-conditioner is 4.0. If it is replaced by another one with the same input power and COP 6.0, the percentage increase in the rate of total thermal energy released to the environment outside is

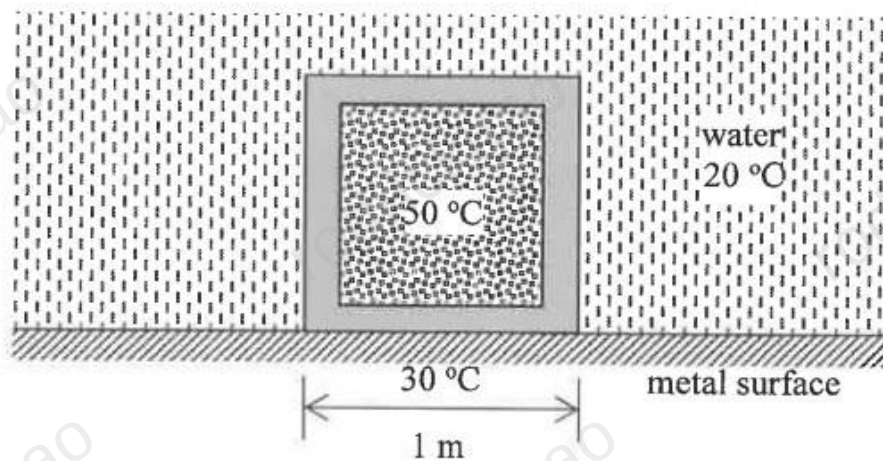
- A. 29%
- B. 33%
- C. 40%
- D. 50%

Official answer: C

Let the unchanged input power be W . Heat rejected outdoors is $Q_H = Q_C + W = (\text{COP} + 1)W$. It rises from $5W$ to $7W$, so the percentage increase is $(7-5)/5 \times 100\% = 40\%$. Answer: C.

3.6

The figure shows a cubical container of side length 1 m submerged in water at 20 °C and its base resting on a metal surface at 30 °C. The container is filled with a liquid at 50 °C. The U-value of the walls is 10 W m⁻² K⁻¹. Find the rate of thermal energy transferred away from the container at the instant shown.



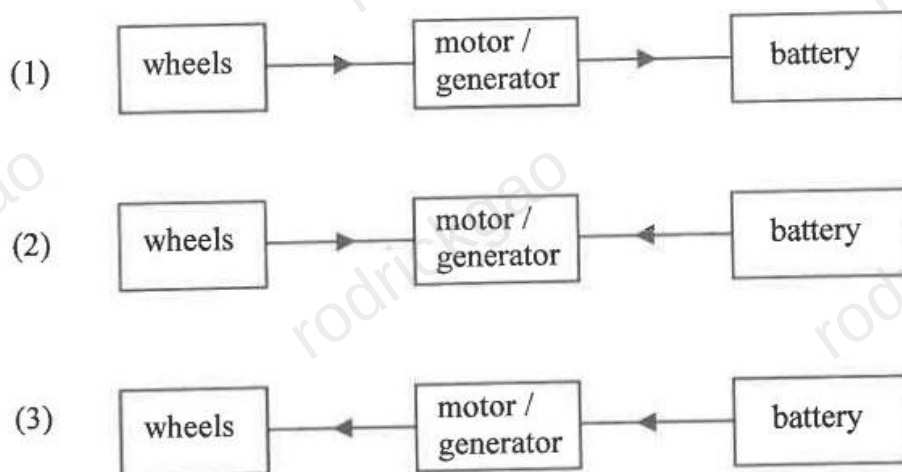
- A. 900 W
- B. 1100 W
- C. 1300 W
- D. 1700 W**

Official answer: D

Five faces, including the top, transfer heat to water with a 30 K temperature difference; the base has a 20 K difference. Each face has area 1 m². Total rate = $UA\Delta T$ summed over the faces = $10[5(1)(50-20)+1(50-30)] = 1700$ W. Answer: D.

3.7

In each schematic diagram, arrows represent the directions of power flow in an electric vehicle's power system. Which diagram most probably corresponds to (i) accelerating and (ii) braking?



- A. accelerating (3); braking (2)
- B. accelerating (3); braking (1)
- C. accelerating (1); braking (2)
- D. accelerating (1); braking (3)

Official answer: B

During acceleration, electrical energy flows from battery to motor to wheels, as in diagram (3). During regenerative braking, the wheels drive the generator and energy returns to the battery, as in diagram (1). Answer: B.

3.8

What factors determine the power generated by a wind turbine?

- (1) air density
- (2) wind speed
- (3) length of the turbine blades

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

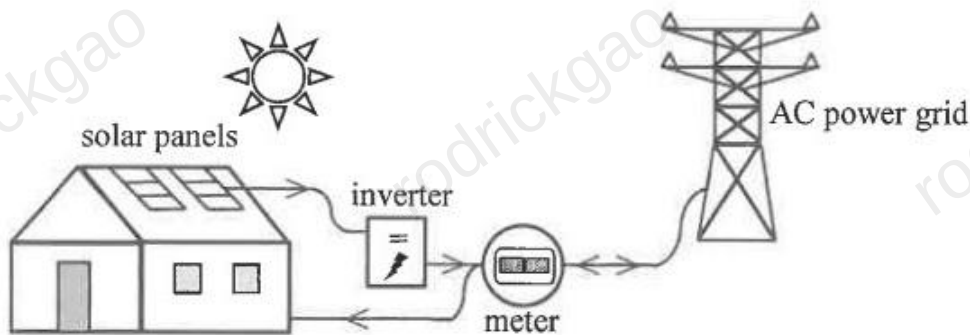
Official answer: D

The available wind power is $P = \frac{1}{2}\rho A v^3$, with swept area $A = \pi r^2$. It therefore depends on air density, wind speed and blade length. All three factors matter. Answer: D.

Question 3: Structured question

Mr Chan plans to install a solar photovoltaic system on the roof of his house for generating electricity domestically.

- (a) Each solar panel has an area of 1.934 m^2 . When it is normal to the Sun, the output voltage and current are 38 V and 10 A respectively on a clear day. Given: maximum solar power per unit area reaching the Earth's surface = 1000 W m^{-2} .
- (i) Find the output power of the solar panel and estimate its efficiency. (2 marks)
- (ii) To achieve a generation capacity of about 10 kW but not exceeding it, find the number of panels to be installed and the corresponding minimum roof area required. (2 marks)
- (b) Mr Chan can take part in the Feed-in Tariff (FiT) Scheme by connecting his solar photovoltaic system to the power grid via an inverter as shown in Figure 3.1. The renewable energy generated can then be sold to the power company.



- (i) Why is an inverter needed between the solar panels and the AC power grid? (1 mark)
- (ii) Throughout a year, Hong Kong has a daily average of 4.5 effective sunshine hours. Estimate the maximum energy, in kW h, that a 10 kW system can deliver in one year. (1 mark)
- (iii) Suggest the main reason why the actual amount of electrical energy generated is far less than the value estimated in (b)(ii). (1 mark)
- (iv) The FiT Scheme offers \$5 per kW h for a system of capacity not exceeding 10 kW, and at most 10 000 kW h can be sold per year. If the initial construction cost is \$200 000 and maintenance costs \$5000 per year, estimate the number of years needed to recover the investment capital. (2 marks)
- (c) State one advantage of solar photovoltaic systems over wind power systems for domestic generation. (1 mark)

Official marking scheme

Solution	Marks	Remarks
3. (a) (i) Output power = $38 \times 10 = 380 \text{ W}$ Input power = $1000 \times 1.934 = 1934 \text{ W}$ Efficiency = $\frac{380}{1934} \times 100\%$ $= 19.648397\% \approx 19.6\%$	1M/1A 1A 2	Accept 19.5 % ~ 20 %
(ii) No. of panels for producing 10 kW $= \frac{10000}{380} = 26.315789 \approx 26$ 26 panels can be installed, i.e. minimum area $= 26 \times 1.934 \text{ m}^2 = 50.284 \text{ m}^2 \approx 50.3 \text{ m}^2$	1M/1A 1A 2	e.c.f. from (a)(i) Accept 50.2 m ² ~ 50.3 m ²
(b) (i) It converts direct current (DC) to alternating current (AC).	1A 1	
(ii) $10 \text{ kW} \times 4.5 \text{ h} \times 365 = 16425 \text{ (kW h)}$	1A 1	Accept 16200 (kW h) ~ 16500 (kW h)
(iii) The solar panel is not always facing the sun. <u>Or The sunlight may be blocked by nearby objects.</u> <u>Or The efficiency of the inverter is not 100%.</u>	1A 1A 1A 1	
(iv) $200000 + 5000 \times t = 5 \times 10000 \times t$ $t = 4.44 \text{ (years)}$	1M 1A 2	Accept 4.4 (years) ~ 4.5 (years)
(c) Solar panels / cells are silent during operation <u>Or lower maintenance cost / installation cost</u> <u>Or comparatively safer as solar panels do not have movable parts</u>	1A 1A 1A 1	

(a)(i) Output power $P = VI = 38(10) = 380 \text{ W}$. Incident power = $1000(1.934) = 1934 \text{ W}$, so efficiency = $380/1934 \times 100\% = 19.6\%$.

(a)(ii) $10000/380 = 26.32$, so install 26 panels to remain below 10 kW. Their capacity is 9.88 kW and their minimum combined area is $26(1.934) = 50.3 \text{ m}^2$.

(b)(i) The inverter converts the panels' direct-current output into alternating current suitable for the AC grid.

(b)(ii) The ideal annual output of the stated 10 kW system is $10(4.5)(365) = 16425 \text{ kW h}$.

(b)(iii) Fixed panels do not face the Sun normally throughout the day, so they receive less solar power than the maximum assumed. Shading or inverter losses is another accepted reason.

(b)(iv) Annual net income = $5(10000) - 5000 = \$45000$. Payback time = $200000/45000 = 4.44 \text{ years}$, using the tariff and sales limit given in the question.

(c) Solar panels operate silently, an advantage for domestic installation.

Section D — Medical Physics

Question 4: Multiple-choice questions

4.1

Sam suffers from short sight and needs spectacles of -1.5 D to correct his far point to the normal position. He breaks them and finds an old pair of spectacles of -1.25 D . How far is he able to see clearly with this pair?

- A. 0.25 m from the eyes
- B. 0.68 m from the eyes
- C. 4 m from the eyes
- D. 7.5 m from the eyes

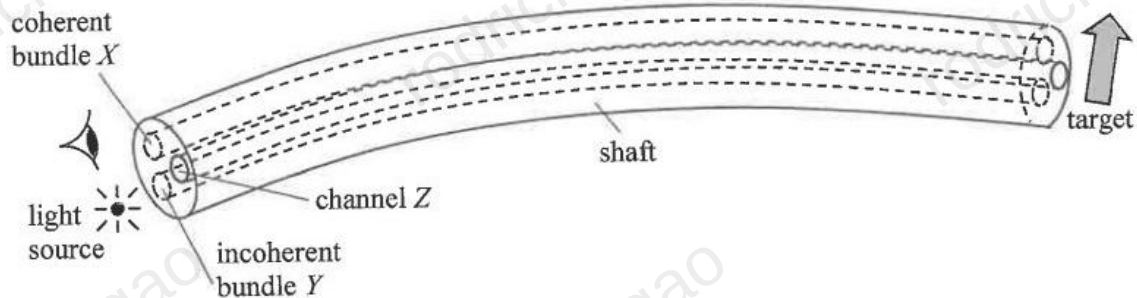
Official answer: C

The correct -1.5 D spectacles form the image of a distant object at the unaided far point, $v = -1/1.5 = -0.667\text{ m}$.
With the -1.25 D pair, $-1.25 = 1/u + 1/v = 1/u - 1.5$, so $u = 4.0\text{ m}$. Answer: C.

4.2

The figure shows a fibre-optic endoscope with two bundles of optical fibres X and Y. Coherent bundle X is for image transmission, while incoherent bundle Y transmits light to illuminate the target. Z is a channel through the shaft. Which statements are correct?

- (1) Bundle X cannot be used to transmit light for illumination.
- (2) Bundle Y cannot be used to transmit images.
- (3) Tools can be inserted through channel Z to take tissue samples for medical testing.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

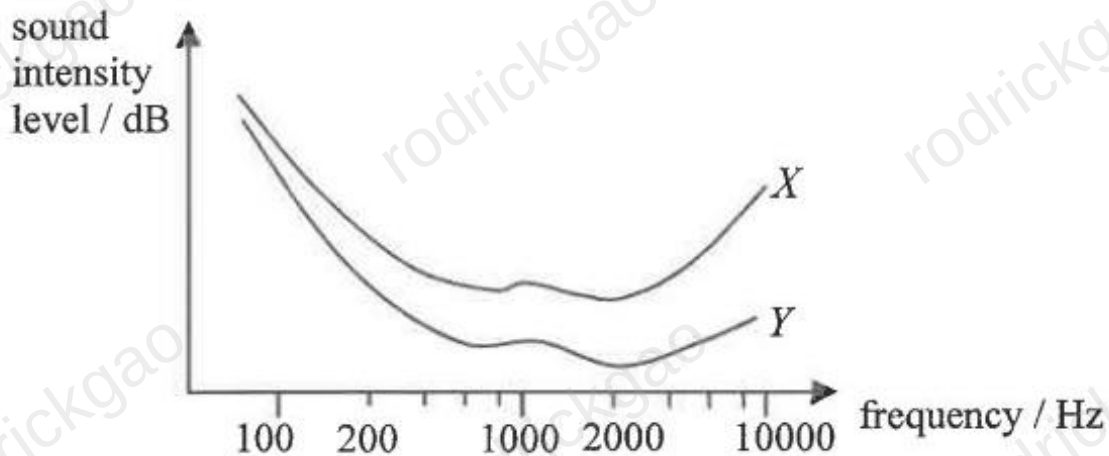
Official answer: D

A coherent bundle preserves the spatial order of fibres and can transmit an image; it can also transmit illumination. An incoherent bundle scrambles image positions and cannot transmit an intact image. The instrument channel admits tools for taking tissue samples. Statements (2) and (3) are correct. Answer: D.

4.3

The figure shows threshold-of-hearing curves of a person with normal hearing and an old man suffering from hearing loss. Which statements are correct?

- (1) Curve Y belongs to the old man.
- (2) The old man's ears are most sensitive to sound at around 2000 Hz.
- (3) The old man's hearing loss is more severe for low-frequency sounds.



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: B

Hearing loss raises the threshold, so upper curve X represents the old man. Its minimum is near 2000 Hz, where his ears are most sensitive. The separation between curves is greater at high frequencies, not low frequencies. Only statement (2) is correct. Answer: B.

4.4

An 8-hour exposure to a sound intensity level of 90.0 dB may cause hearing loss. How much energy falls on an eardrum of area 0.503 cm^2 for such exposure? Given: intensity at the threshold of hearing = $10^{-12} \text{ W m}^{-2}$.

- A. 1.45 J
- B. $1.45 \times 10^{-3} \text{ J}$
- C. $1.81 \times 10^{-4} \text{ J}$
- D. $2.42 \times 10^{-5} \text{ J}$

Official answer: B

From $L = 10 \log_{10}(I/I_0)$, $I = 10^{-12} \times 10^9 = 10^{-3} \text{ W m}^{-2}$. Convert the area to $0.503 \times 10^{-4} \text{ m}^2$ and the time to $8 \times 3600 \text{ s}$. Energy = $IAt = (10^{-3})(0.503 \times 10^{-4})(28800) = 1.45 \times 10^{-3} \text{ J}$. Answer: B.

4.5

Which statement about the acoustic impedance of a material to ultrasound is correct?

- A. It represents how much resistance an ultrasound beam encounters as the beam passes through the material.
- B. Its value increases with the mass of the material.
- C. Its value depends on the frequency of the ultrasound.
- D. Perfect reflection occurs at a boundary between two tissues having the same acoustic impedance.

Official answer: A

Acoustic impedance $Z = \rho c$ characterizes a medium's opposition to oscillatory particle motion; for a travelling wave it relates acoustic pressure to particle velocity. It depends on material properties rather than total mass. Equal impedances give zero reflection at an ideal boundary, not perfect reflection. Answer: A.

4.6

When an ultrasound scanner is used to examine a large organ, the ultrasound must reach locations deep inside the body. Which can help achieve this purpose?

- (1) applying coupling gel to fill the air gap between transducer and skin
- (2) using a transducer employing lower-frequency ultrasound
- (3) using a transducer with higher resolving power

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Coupling gel removes the highly reflecting air gap, allowing more ultrasound to enter the body. Lower-frequency ultrasound is attenuated less and penetrates more deeply. Higher resolving power usually requires a higher frequency, which reduces penetration. Statements (1) and (2) are correct. Answer: A.

4.7

The linear attenuation coefficients of soft tissue for X-rays of 20 keV and 60 keV are 0.77 cm^{-1} and 0.21 cm^{-1} respectively. Determine the intensity ratio of 20 keV to 60 keV X-ray beams transmitted respectively through 20 cm of soft tissue. Both beams have the same initial intensity.

- A. 2.1×10^{-7}
- B. 7.1×10^{-6}
- C. 1.4×10^{-5}
- D. 1.5×10^{-2}

Official answer: C

With equal incident intensities, $I_{20}/I_{60} = \exp[-(\mu_{20}-\mu_{60})x] = \exp[-(0.77-0.21)(20)] = 1.37 \times 10^{-5} \approx 1.4 \times 10^{-5}$.
Keep thickness in centimetres to match the attenuation coefficients. Answer: C.

4.8

The typical total scan and image-processing time for computed tomography (CT), radionuclide imaging (RNI) and X-ray radiography (XP), in ascending order, is

- A. $\text{RNI} < \text{CT} < \text{XP}$
- B. $\text{XP} < \text{RNI} < \text{CT}$
- C. $\text{CT} < \text{XP} < \text{RNI}$
- D. $\text{XP} < \text{CT} < \text{RNI}$

Official answer: D

For the typical procedures compared in this question, a plain X-ray image is quickest, CT requires multiple projections and reconstruction, and radionuclide imaging usually requires longer count collection. The order is $\text{XP} < \text{CT} < \text{RNI}$. Answer: D.

Question 4: Structured question

(a) The table below shows some information about air, skin and muscle.

Material	Density / kg m^{-3}	Speed of sound / m s^{-1}
air	1.20	340
skin	1000	1520
muscle	1040	1630

- Find the acoustic impedance of muscle. (1 mark)
 - An ultrasound transducer is inclined at 5° to the normal of the skin surface when used for a scan. Find the angle of refraction of the ultrasound beam after it enters the skin from air. (2 marks)
 - Hence explain why an ultrasound transducer should be held perpendicular to the skin surface during a scan. (2 marks)
- (b) Figure 4.1(a) is a radionuclide image showing a patient's bone scan. Figure 4.1(b) is a chest X-ray image.



Figure 4.1(a)



Figure 4.1(b)

- In terms of the nature of the radiation source and the image-production mechanism, compare how the images in Figures 4.1(a) and 4.1(b) are produced. There is no need to mention the detection instruments or their detecting mechanisms. (3 marks)
- Radionuclide imaging can provide information that X-ray radiographic imaging cannot. Explain briefly. (2 marks)

Official marking scheme

Solution	Marks	Remarks
4. (a) (i) acoustic impedance = density $\times$ speed $= 1040 \times 1630$ $= 1.6952 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$ $\approx 1.70 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$ (Rayl)	1A 1	
(ii) $\frac{\sin \theta_{\text{air}}}{\sin \theta_{\text{skin}}} = \frac{\text{speed of sound in air}}{\text{speed of sound in skin}} = \frac{340}{1520}$ $\sin \theta_{\text{skin}} = \frac{1520}{340} \times \sin 5^\circ = 0.39$ $\Rightarrow \theta_{\text{skin}} = 22.93^\circ \approx 22.9^\circ$	1M 1A 2	
(iii) Refraction distortions occur when the ultrasound beam is bent from its original direction as it passes through a boundary between tissues having different sound speeds. (One would assume that the beam goes straight in ultrasound scans and does not know that the sound path has been altered due to refraction.) It results in improper positioning and/or brightness of echoes displayed in ultrasound images.	1A 1A 2	
(b) (i) Radionuclide image (bone scan) – radioactive γ source (radio-pharmaceutical) is injected to the body and carried to target organs. Different concentration of γ source in the body gives different brightness in image. Chest X-rays – X-rays are produced by an X-ray tube. When X-rays pass through the body from outside, they are absorbed by different tissues. The different attenuation of X-rays gives different brightness in image.	1A 1A 1A 3	
(ii) (Explain in terms of selective uptake of radionuclide and functional information of the organ.) Radio-pharmaceutical targets the bone / organ due to selective uptake as (bone) hot spots. Hence bone scan / RNI detects function (or physiology) due to the metabolic uptake of the radio-pharmaceutical rather than anatomy as do X-rays. It helps to locate where the cancer has been spread.	1A 1A 2	

(a)(i) $Z = \rho c = 1040(1630) = 1.70 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$.

(a)(ii) Apply refraction of sound: $\sin \theta_{\text{air}}/c_{\text{air}} = \sin \theta_{\text{skin}}/c_{\text{skin}}$. Thus $\sin \theta_{\text{skin}} = (1520/340)\sin 5^\circ$ and $\theta_{\text{skin}} = 22.9^\circ$.

(a)(iii) Refraction changes the beam's path, while the scanner assumes approximately straight propagation. This can misplace structures and alter displayed echoes. Holding the transducer normal to the skin avoids bending at that boundary and reduces such distortion.

(b)(i) In a bone scan, an administered radiopharmaceutical is selectively taken up by tissues and emits gamma rays from inside the body. Differences in uptake and activity produce the image's brightness pattern. In chest radiography, an external X-ray tube sends X-rays through the body; tissues attenuate the beam by different amounts, producing the image contrast.

(b)(ii) Selective tracer uptake reflects metabolic activity and organ function, while an ordinary radiograph primarily shows anatomy through attenuation. Abnormal uptake can reveal bone sites affected by cancer spread even when structural changes are not yet apparent on a plain radiograph.